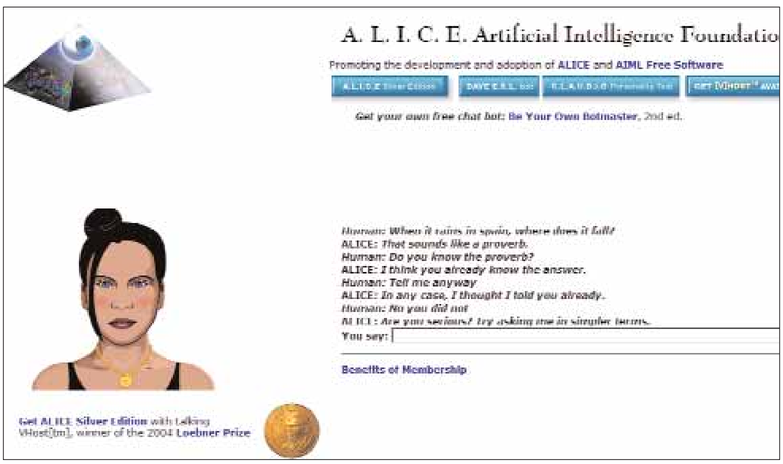
**For now, we just have to make do with Word auto-correcting our obvious typoes**

We've spoken about the need for better interfaces in this *Sixth Sense* section, and there's a dire need to move away from the keyboard and mouse system of input. But how do we achieve that without smarter software? Here lies the problem. Artificial Intelligence (AI) has been a dream for over half a century now. Researchers have been building robots and software to try and mimic the human brain for as long as they possibly could, and are nowhere near building a thinking application than they were when they started. A lot of people will disagree with that, of course, and we don't mind. A.L.I.C.E. (Artificial Linguistic Internet Computer Entity), **www.alicebot.org**, can simulate intelligence to an untrained observer, but doesn't *understand* the meaning of what's being said. Basically, everything to do with computers uses mathematics, and it's near-impossible to make a computer understand characteristics of human speech or conversation such as wit and sarcasm. And if that's hard, imagine how impossible it is for a computer to understand the difference between someone telling the truth and someone lying!




As you can see from this chat, Alice tends to lose her way sometimes

The beauty of human intelligence springs from its complexity, but forget us superior humans. What about dogs? How do we teach a robotic dog to have instincts? Will a robotic dog ever be able to "smell fear"? Will we ever be able to make a robotic dog that knows when you're happy and wags its tail? Until we fully understand and can translate into mathematics things like instinct, feelings, and sarcasm, it's going to be near-impossible to attain true AI.

AI Research (**www.a-i.com**) is one of the leading AI organisations in the world, and is taking a step backwards—instead of trying to make an adult brain, it's focussing on making an artificial child. The idea is that a child's brain is much less complex than an adult's, and like a child, we need our AI to learn. We need to teach our AI the same way we all learnt as children—how to speak, nursery rhymes,... and gradually build language. This where we get our understanding of the use of language, and hopefully AI Research's child will do so too. It's going to take a long time though, but perhaps this is the right method.

We're getting ahead of ourselves here. Sticking to better software, and AI in software, it's acceptable to have simulated intelligence, because we just want the software to help us work faster and better. Think about something as simple as AutoCorrect options in MS Word; if it can actually read and *understand* the sentences we type: perhaps someday MS Word will actually see you type "News of his father's death sent him into sock," understand that the word "death" is something bad and sorrowful, "father" is a very close relative, and no one can really "go into a sock." It will then see that the word should have been "shock" and not "sock"—or for that matter "sick", "sack", or "stock". Complicated, but it doesn't sound impossible to us. For now, we just have to make do with being auto-corrected for obvious typoes.

With any single-player game, where the computer controls other entities, AI is used. The problem is that the human brain is advanced enough to figure out behavioural patterns of AI. Because it's based on mathematics, AI can get quite predictable. We humans are more chaotic, and especially when playing a game, no two humans have the same patterns or behaviours. In fact, the same person playing the same game for the second time might play it differently from the first time. With CPUs getting faster by the minute, the future will not be held back by lack of computing power—it's now up to the developers of games and AI to harness that power and make AI players in games a lot more believable.